Modelling Choice and Reaction Time in Delay Discounting with an Absolute-Conflict RT Model

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This report presents my part of a three-person group project for the course Computational Cognitive Science (Summer Term 2026).

1 OVERVIEW

This report analyses choice and reaction time (RT) data from the Delay Discounting Data Set, collected from 99 participants. Subjective values follow a hyperbolic discounting model, choices follow a softmax rule on the value difference ΔV, and RT follows a shifted log-normal distribution whose location depends on $|\Delta V|$ (the absolute-conflict model). We assess parameter recovery, test-retest reliability across two runs, and out-of-sample prediction, and we test whether adding RT improves on a choice-only baseline. The original group project compared three RT models. This report covers only the absolute-conflict model and the choice-only baseline.

2 METHODS

2.1 Data

We used the Delay Discounting Data Set obtained from 99 participants performing a delay discounting task, abbreviated as the DD data set. It was sampled from two experiments with slightly different setups: 49 participants (Sample-1) and 50 participants (Sample-2). Each trial records the immediate and delayed reward magnitudes (r_{imm}, r_{del}), the delay (days), the choice (1 = immediate, 2 = delayed, 0 = missing), an experimental condition parameter ($p_{imm} \in [0, 1]$), the condition (1 = reward discounting, 2 = loss discounting), and reaction time (RT; ms). Based on the first three variables listed above (r_{imm}, r_{del}, and delay), we compute each option's subjective value via the valuation model (Section 2.2); the resulting value difference ΔV is the shared predictor driving both choice and RT. The condition column takes only the value 1 (reward discounting) throughout the DD data set, meaning the loss condition contributes no data. Before preprocessing, run A consists of 80 trials from Sample-1 and 100 trials from Sample-2 per participant, whereas run B consists of 180 trials from Sample-1 and 225 trials from Sample-2 per participant.

Preprocessing (per subject, per run, in order): (1) drop any trial with a missing value (NaN) in any of the seven raw columns; (2) drop missing-choice trials (choice $= 0$); (3) drop RT < 200 ms; (4) drop RT $> Q3 + 3 \times$ IQR (IQR computed per subject/run on the data remaining after steps 1–3). Across all 99 subjects and both runs: 28,990 trials before cleaning; 152 removed for missing values (0.52%), which subsume all 93 missing-choice trials (0.32%); 0 for RT < 200 ms; 363 for RT outliers (1.25%). In total, 28,475 trials remain, a removal rate of 1.78%, with per-subject removal ranging 0–9.6%.

In run A, 45.2% of trials ($n = 8794$) had a delayed choice, compared to 60.0% in run B ($n = 19681$). For RT, under the (only) reward condition, mean (standard deviation, SD) RT was 1877.4 (811.3) ms in run A and 1664.1 (816.7) ms in run B; medians were 1707.7 ms and 1493.9 ms, respectively. Since the mean exceeds the median in both runs, the RT distribution is right-skewed.

2.2 Models

Valuation Model and Choice Rule. We adopt the classical hyperbolic discounting model that assigns a subjective value to each option: $V = \dfrac{A}{1 + kD}$, where A takes r_{imm} or r_{del} as the reward magnitude, and D is the delay in days, with $D = 0$ for the immediate option. The discounting rate $k > 0$ is estimated per participant; a larger k indicates a stronger preference for immediate over delayed reward.

We then compute the value difference between the two options, $\Delta V = V_{delayed} - V_{immediate}$, and couple it to choice via a softmax/logistic rule,

$$P(\text{delayed}) = \text{sigmoid}(\beta\,\Delta V) = \frac{1}{1+\exp(-\beta\,\Delta V)},$$

where $\beta > 0$ is estimated; a larger β indicates greater sensitivity to ΔV. For trial i, the choice likelihood is

$$p(\text{choice}_i \mid \Delta V_i, \beta) = \text{sigmoid}(\beta\,\Delta V_i)^{c_i}\,\big(1 - \text{sigmoid}(\beta\,\Delta V_i)\big)^{1-c_i}, \tag{1}$$

where $c_i = 1$ if the choice was delayed and $c_i = 0$ if immediate.

RT model. We assume RT $\sim$ shifted log-normal, i.e. $\log(\text{RT} - \text{shift}) \sim \mathcal{N}(\mu_i, \sigma)$. The shift for each participant is fixed as shift $= 0.8 \times \min(\text{RT})$, computed once from the cleaned RT pooled across both runs and reused for both runs' likelihood computations. This guarantees RT $-$ shift > 0 in all trials.

For trial i, the RT likelihood follows the shifted log-normal density,

$$p(\text{RT}_i \mid \mu_i, \sigma, \text{shift}) = \frac{1}{(\text{RT}_i - \text{shift})\,\sigma\sqrt{2\pi}} \exp\left(-\frac{\big(\log(\text{RT}_i - \text{shift}) - \mu_i\big)^2}{2\sigma^2}\right). \tag{2}$$

Combining Eq. (1) and Eq. (2), the full model's joint trial likelihood is

$$p(\text{choice}_i, \text{RT}_i \mid \theta) = p(\text{choice}_i \mid \Delta V_i, \beta)\; p(\text{RT}_i \mid \mu_i, \sigma, \text{shift}), \qquad \theta = \{k, \beta, \theta_\mu, \sigma\},$$

where $\theta_\mu = \{a, b\}$ denotes the parameters of the μ mapping defined below.

Absolute-conflict model. Response time is governed by decision difficulty, quantified as the absolute value difference $|\Delta V_i|$: response time is longer as $|\Delta V_i|$ approaches 0. Accordingly, $\mu_i = a - b|\Delta V_i|$: the slowdown scales linearly with $|\Delta V_i|$, where a is the RT baseline and b denotes the sensitivity to conflict [1].

Choice-only baseline. The choice-only baseline uses the identical valuation model and choice rule, with parameters $\theta = \{k, \beta\}$ estimated from the choice likelihood Eq. (1) alone, i.e. the RT term is dropped. Both k and β are fit independently in the baseline and in the full model, making the two directly comparable, so that the effect of adding RT can be observed.

2.3 Inference

To keep the optimiser from diverging while leaving parameters largely unconstrained by the data, we use weak priors and bounds: $k \sim \text{HalfNormal}(\text{scale} = 1)$ with $k \in (10^{-6}, 10)$, $\beta \sim \text{HalfNormal}(5)$ with $\beta \in (10^{-6}, 50)$, $\sigma \sim \text{HalfNormal}(1)$ with $\sigma \in (10^{-6}, 5)$, $a \sim \mathcal{N}(\log(\text{median RT of the data being fitted}), 2)$, unbounded, and $b \sim \text{HalfNormal}(2)$ with $b \in [0, 20]$. The baseline uses the k, β priors only.

We implement per-participant maximum a posteriori (MAP) estimation as the objective function:

$$\hat{\theta} = \arg\min_{\theta} -\big[\log p(\text{choice}, \text{RT} \mid \theta) + \log p(\theta)\big],$$

using `scipy.optimize.minimize` with method L-BFGS-B.

Every per-participant fit (run A, run B, and the parameter-recovery refits) uses a 6-candidate pool: one heuristic single start plus 5 random starts drawn uniformly within each bounded parameter's range (except a, drawn from its prior). We keep whichever of the six candidates attains the lowest negative log posterior. The heuristic start is $k = 0.05, \beta = 1, \sigma = 0.5$ for the shared parameters and $a = \log(\text{median RT}), b = 0.1$ for θ_μ.

2.4 Evaluation Plan

(i) Parameter recovery. For each participant, we simulate choice and RT data using the same trial design (r_{imm}, r_{del}, and delay) with that participant's run-A MAP estimate as ground truth, refit with the identical pipeline, and correlate true vs. recovered parameters via Pearson r. The same simulated choices are also refit with the choice-only baseline, to test whether RT sharpens identifiability of k, β.

(ii) Reliability. We fit run A and run B independently using the same pipeline but distinct random-start seeds, and compute Pearson r and ICC(2,1) (two-way random effects, absolute agreement)[2] between the two runs' parameter estimates across participants, in both the full model and the baseline.

(iii) Out-of-sample prediction. We apply each subject's run-A-fitted parameters to run-B data and report, for both models, choice predictive log-likelihood,

$$\ell_{\text{choice}} = \frac{1}{N_B} \sum_{i=1}^{N_B} \log p(\text{choice}_i \mid \Delta V_i, \hat{\beta}),$$

and accuracy; for the full model only, we additionally report RT predictive log-likelihood,

$$\ell_{\text{RT}} = \frac{1}{N_B} \sum_{i=1}^{N_B} \log p(\text{RT}_i \mid \mu_i, \hat{\sigma}, \text{shift}),$$

where N_B is the total number of run-B trials pooled across all subjects. All metrics are computed under the reward condition only (Section 2.1).

(iv) Value of RT. The full model is compared against the choice-only baseline on the choice predictive quantity from (iii), isolating whether adding RT sharpens the choice account. Whether RT sharpens identifiability and reliability of k, β is assessed via the full-vs-baseline comparisons already described in (i) and (ii).

3 RESULTS

3.1 Parameter Recovery and Identifiability

Table 1 shows that k, a, and σ recover well ($r \geq 0.90$), β recovers similarly well in both the full model and the baseline ($r = 0.898$ and $r = 0.899$), while b recovers more weakly ($r = 0.758$). This is not a boundary artifact: excluding the 15/99 synthetic subjects whose optimiser attained the $b \geq 0$ lower bound, the remaining 84 subjects recover even more weakly at $r = 0.711$. The weaker identifiability of b instead reflects a genuinely small conflict effect across all 99 participants: the conflict term $b \times \text{median}(|\Delta V|)/a$ has a median of just 0.26% (max 1.1%), indicating b has little influence on the RT likelihood even under correct model specification. Adding RT changes the identifiability of k and β only marginally relative to the baseline ($\Delta r_k = +0.013$, $\Delta r_\beta = -0.001$), i.e. modelling RT neither substantially sharpens nor degrades their recovery.

Parameter	r (full model)	r (baseline)
k	0.904	0.891
β	0.898	0.899
a	0.966	—
b	0.758	—
σ	0.935	—

Table 1: Parameter recovery (Pearson r), absolute-conflict model vs. choice-only baseline.

3.2 Reliability

Table 2 shows that test-retest reliability is low throughout, and, notably, the full model's k is less reliable than the baseline's k, despite both being fit to run A/B data generated by the same subject. This gap is not an artifact of a handful of outliers: excluding the 5 most discrepant subjects raises r_k from 0.091 to 0.396 in the full model and from 0.248 to 0.680 in the baseline, so the gap widens rather than narrows (from -0.157 to -0.284). We also ruled out cross-parameter entanglement (all $|r| < 0.18$ between k and $\{\beta, a, b, \sigma\}$ within a run) and a full-model identifiability defect, since k's parameter recovery is if anything better for the full model. We instead attribute the gap to per-participant (non-hierarchical) MAP estimation under a modest per-subject trial count, compounded by a small conflict effect: the full model estimates three additional parameters (a, b, σ) without gaining much additional constraining information from RT, making its 5-dimensional posterior harder to pin down consistently across two independently fitted runs than the baseline's.

Model	Parameter	Pearson r	ICC(2,1)
Full	k	0.091	0.089
Full	β	0.392	0.382
Full	a	0.432	0.311
Full	b	0.229	0.035
Full	σ	0.261	0.206
Baseline	k	0.248	0.249
Baseline	β	0.409	0.396

Table 2: Run A vs. run B reliability (Pearson r and ICC(2,1)), absolute-conflict model vs. choice-only baseline.

3.3 Predictive Performance

As noted in Section 2.1, loss-condition trials do not occur in this data set. Table 3 summarises out-of-sample prediction on run B, using run-A-fitted parameters. For the absolute-conflict model, the choice-only baseline outperforms the full model on both choice predictive log-likelihood and accuracy. For both models, the choice log-likelihood per trial (-1.679 and -1.481) is below the chance level of $\log 0.5 \approx -0.693$, although both models reach a choice accuracy above 0.68. The RT log-likelihood has no comparison model in this report and is listed for completeness only.

Model	Choice log-lik/trial	Choice accuracy	RT log-lik/trial
Full: Absolute Conflict	-1.679	0.681	-8.397
Baseline	-1.481	0.686	–

Table 3: Out-of-sample predictive performance on run B (reward condition; $n = 99$ subjects, 19,681 trials). RT log-likelihood is not defined for the choice-only baseline.

3.4 Value of the RT Modality

For the absolute-conflict model, adding RT slightly lowers out-of-sample choice prediction: pooled choice log-likelihood is worse (-1.679 vs. -1.481) and pooled accuracy is marginally lower (0.681 vs. 0.686; Table 3). Two subjects, whose run-A β estimates (6.403 and 4.566) lie far above the median of 0.719, account for 99.0% of the summed run-B choice log-likelihood difference (3858.0 of 3896.2). Subject by subject, the baseline is better for 54 and the full model for 45 of the 99 subjects. RT also fails to sharpen identifiability: k, β recovery changes only marginally versus the baseline ($\Delta r_k = +0.013$, $\Delta r_\beta = -0.001$; Section 3.1). Modelling RT thus adds parameters and complexity without a compensating gain in choice prediction or parameter stability.

4 CONCLUSION

For the absolute-conflict model, k, β, a, and σ are well recovered, while b recovers more weakly because the conflict effect is small. Test-retest reliability is low, and the full model's k is less reliable than the baseline's k. Adding RT slightly lowers out-of-sample choice prediction and changes the recovery of k and β only marginally.

REFERENCES

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